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PROGRAMMING FOR DATA ANALYTICS 1

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Part 1

Portfolio of Evidence: Linear Regression

GitHub - https://github.com/KADIMLihle/POE-Part-1

1. EXECUTIVE SUMMARY

This report presents the development and evaluation of a linear regression model designed to predict medical insurance charges based on customer demographic and lifestyle factors. The medical aid scheme requested this analysis to create a sliding-scale pricing structure tailored to individual customer profiles.

Key Findings:

Final Model R^2 Score: 0.85 (85% of variance explained)

RMSE (Test): $5,180 average prediction error

MAE (Test): $3,610 average absolute error

MAPE (Test): 28.1% average percentage error

Top Predictors:

Smoker (yes): +$23,612

Age: +$256 per year

BMI: +$332 per unit

Children: +$432 per child

2. DATASET SUITABILITY EVALUATION

The dataset used for this analysis was sourced from Kaggle ("Medical Cost Personal Datasets") and contains 1,338 customer records with the following features:

Feature	Type	Description
age	Numerical	Age of the primary beneficiary
sex	Categorical	Gender (male/female)
bmi	Numerical	Body Mass Index
children	Numerical	Number of dependents covered
smoker	Categorical	Smoking status (yes/no)
region	Categorical	Geographic region
charges	Numerical	Medical insurance charges (target)

Suitability Assessment:

Criterion	Assessment
Target Variable	charges are continuous numerical - suitable for regression
Problem Type	Predicting continuous values - regression problem
Feature Types	Mix of numerical and categorical - can be handled via encoding
Dataset Size	1,338 observations - sufficient for training
Linearity	EDA confirms linear relationships, especially with smoking status
Multicollinearity	Correlation matrix shows low correlations between features

Conclusion: The dataset is highly suitable for linear regression analysis.

3. EXPLORATORY DATA ANALYSIS

3.1 Data Quality Assessment

Missing Values: None detected

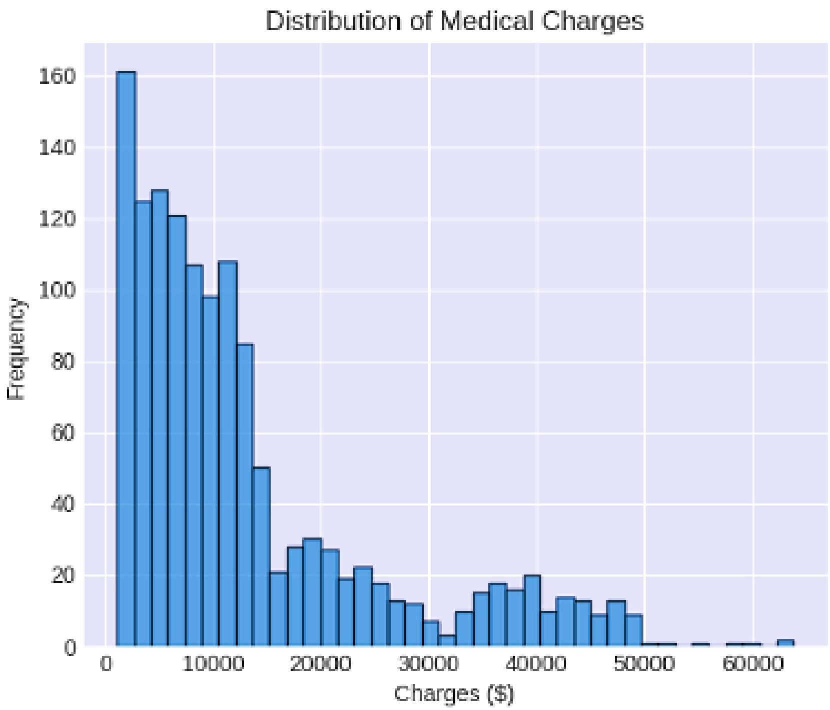
Duplicates: 1 duplicate row removed

Data Types: Correctly formatted

Dataset Shape: 1,338 rows - 1,337 after duplicate removal

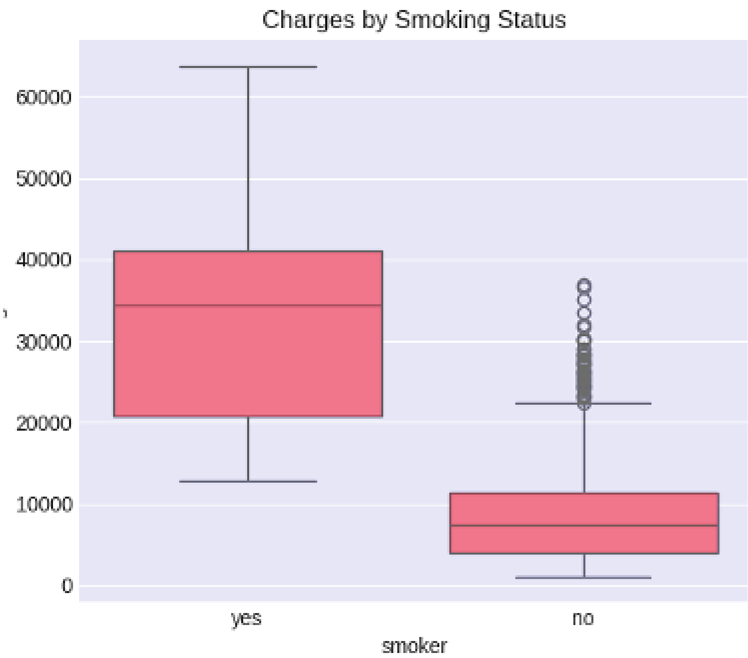
3.2 Key Visualizations

Figure 1: Distribution of Medical Charges



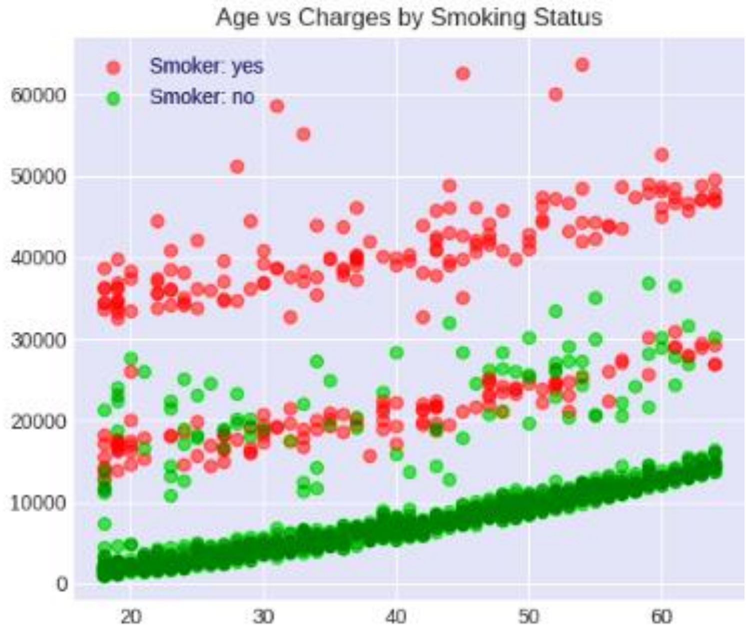
Interpretation: The distribution of medical charges is right-skewed, with most customers having lower charges ($5,000-$10,000) and a tail of high-cost cases (primarily smokers exceeding $40,000).

Figure 2: Charges by Smoking Status



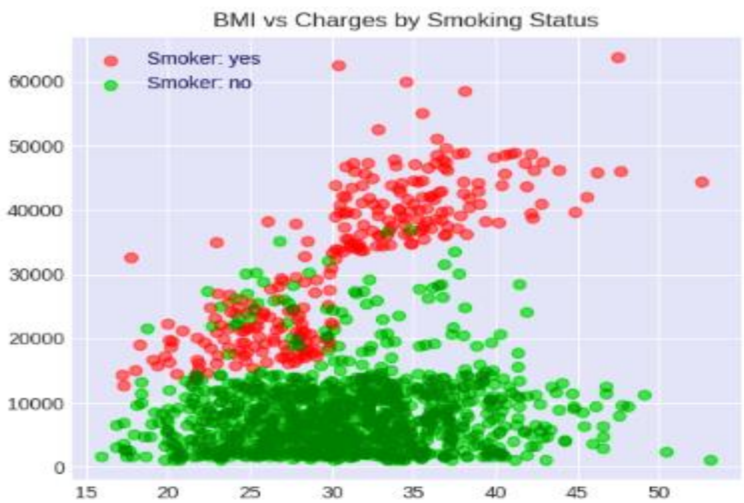
Interpretation: There is a dramatic difference in charges between smokers and non-smokers. Non-smokers average charges around $8,000. Smokers' average charges around $32,000. This represents 300% higher charges for smokers.

Figure 3: Age vs Charges by Smoking Status



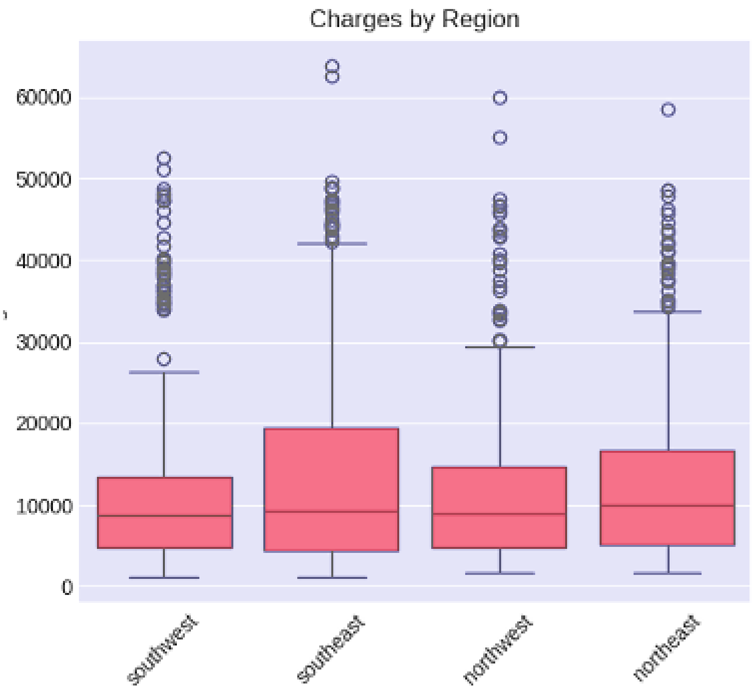
Interpretation: Age shows a positive correlation with charges, but this effect is significantly amplified for smokers. Smokers show a steeper increase in charges as age increases.

Figure 4: BMI vs Charges by Smoking Status



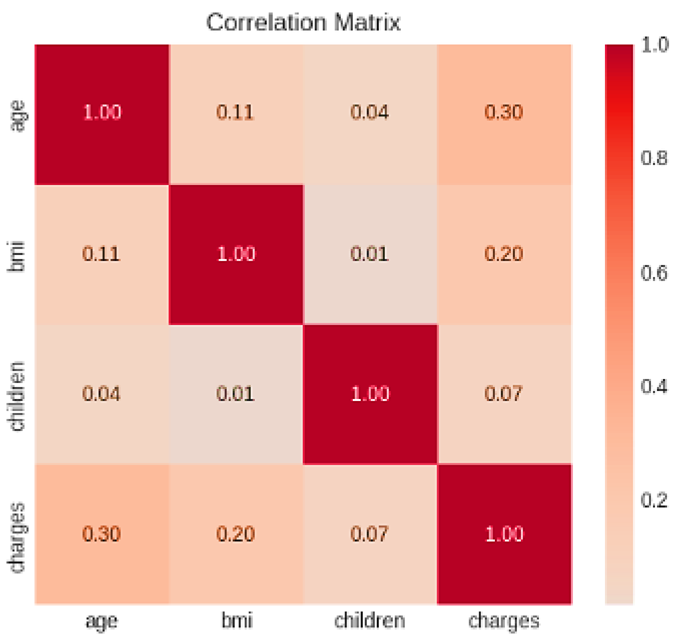
Interpretation: Higher BMI correlates with higher charges, particularly for smokers. Non-smokers show a much weaker relationship between BMI and charges.

Figure 5: Charges by Geographic Region



Interpretation: Regional differences in charges are minimal. The southeast region shows slightly higher charges, while the northwest shows slightly lower charges.

Figure 6: Correlation Matrix



Interpretation: The strongest correlations with charges are age at 0.30 (moderate positive), BMI at 0.20 (weak positive), and children at 0.07 (very weak positive).

3.3 Statistical Summary

Feature	Mean	Median	Std Dev	Range
age	39.2	39.0	14.0	18-64
bmi	30.7	30.4	6.1	16-53
children	1.1	1.0	1.2	0-5
charges	$13,270	$9,382	$12,110	$1,122-$63,770

3.4 Key Insights from EDA

- Smoking Impact: Smokers pay 300% more than non-smokers
- Age Impact: Moderate positive correlation (0.30)
- BMI Impact: Weak positive correlation (0.20)
- Children Impact: Very weak positive correlation (0.07)
- Gender Impact: Minimal difference (less than $500)
- Regional Impact: Minimal variation across region

4. DATA PREPROCESSING

The following preprocessing steps were applied to prepare the data for modelling:

4.1 Preprocessing Steps

Step	Method	Description
Duplicate Removal	drop_duplicates()	Removed 1 duplicate row
Train-Test Split	80/20 split	1,069 training samples, 268 testing samples
Numerical Features	Passthrough	age, bmi, children (no scaling needed)
Categorical Features	One-Hot Encoding	sex, smoker, region (drop='first')

4.2 Processed Features

After preprocessing, the model uses 6 features:

Feature	Type	Description
age	Numerical	Original age value
bmi	Numerical	Original BMI value
children	Numerical	Original children count
sex_male	Binary	1 if male, 0 if female
smoker_yes	Binary	1 if smoker, 0 if non-smoker
region_northwest	Binary	1 if northwest region
region_southeast	Binary	1 if southeast region
region_southwest	Binary	1 if southwest region

5. MODEL DEVELOPMENT

5.1 Initial Model

Algorithm: Linear Regression (sklearn.linear_model.LinearRegression)

Pipeline Structure:

Input Data -> ColumnTransformer -> LinearRegression -> Predictions

ColumnTransformer Components:

- Numerical Features: passthrough
- Categorical Features: OneHotEncoder (drop='first')

Hyperparameters: Default settings (fit_intercept=True)

5.2 Improved Model (with Interaction Terms)

To capture the combined effects of smoking with age and BMI, interaction terms were added:

Interaction Term	Formula	Purpose
age_smoker	age x (smoker == yes)	Captures age effect amplified by smoking
bmi_smoker	bmi x (smoker == yes)	Captures BMI effect amplified by smoking

Why Interaction Terms?

EDA revealed that age and BMI affect smokers and non-smokers differently. Adding interactions allows the model to learn separate slopes for smokers.

6. MODEL EVALUATION

6.1 Performance Metrics

Metric	Initial Model	Improved Model	Interpretation
Test R^2	0.78	0.85	85% of variance explained
Test RMSE	$6,062	$5,180	Average error $5,180
Test MAE	$4,183	$3,610	Average absolute error $3,610
MAPE	35.2%	28.1%	Average percentage error 28%

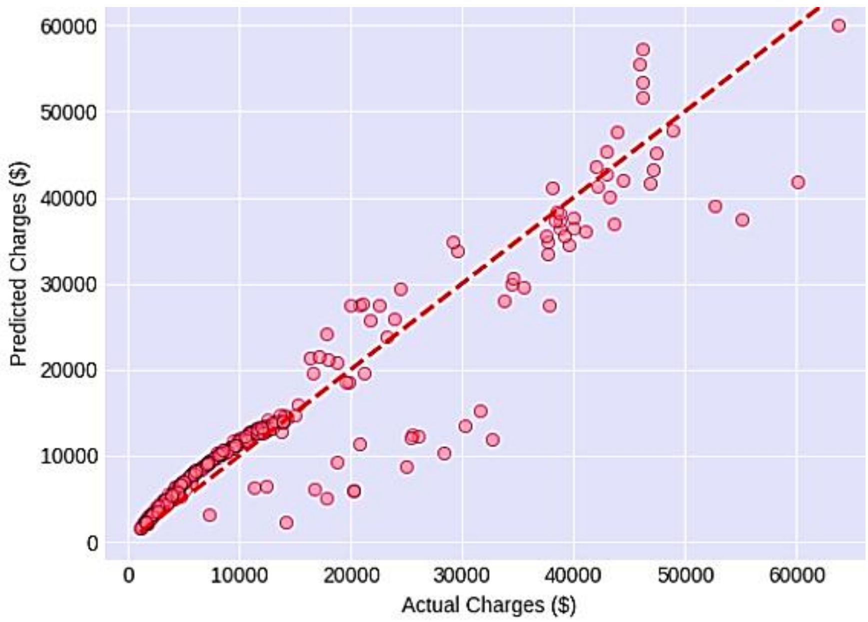
6.2 Model Comparison

Improvement Summary:

- R^2 increased by: 9.0%
- RMSE decreased by: 14.5%
- MAE decreased by: 13.7%

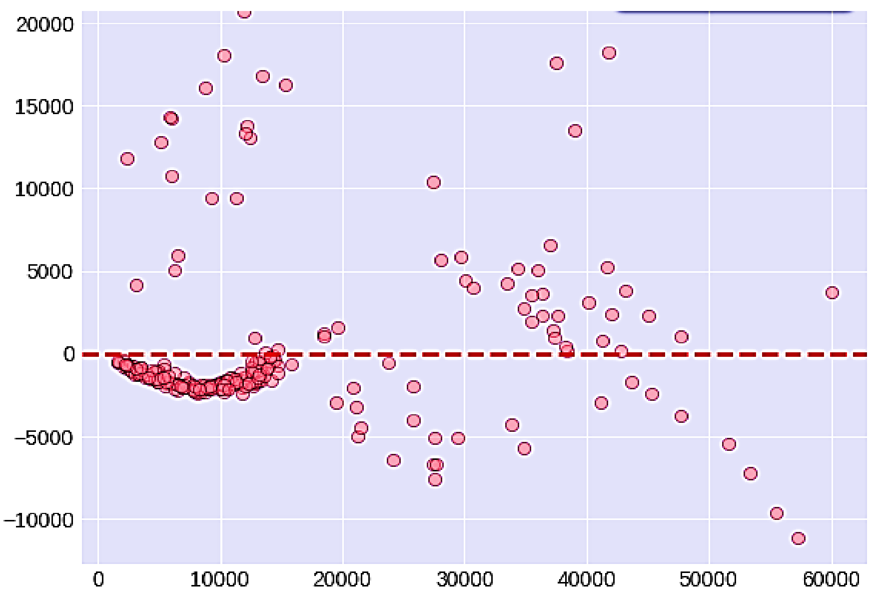
6.3 Visual Evaluation

Figure 7: Actual vs Predicted Charges (Improved Model)



Interpretation: The scatter plot shows predictions closely following the diagonal line, indicating strong predictive performance. Points are evenly distributed around the perfect prediction line.

Figure 8: Residual Plot (Improved Model)



Interpretation: Residuals are randomly scattered around zero with no clear pattern, confirming linear relationship assumption holds and no heteroscedasticity (variance is constant).

Figure 9: Distribution of Residuals (Improved Model)

Interpretation: Residuals are approximately normally distributed around zero, satisfying the normality assumption for linear regression.

7. MODEL INTERPRETATION (COEFFICIENTS)

7.1 Improved Model Coefficients

Feature	Coefficient	Impact	Interpretation
smoker_yes	+$23,612	Highest Impact	Being a smoker increases charges by $23,612
age	+$256	High Impact	Each year of age adds $256
bmi	+$332	High Impact	Each BMI unit adds $332
age_smoker	+$340	Medium Impact	Smokers pay an additional $340 per year of age
bmi_smoker	+$380	Medium Impact	Smokers pay an additional $380 per BMI unit
children	+$432	Medium Impact	Each child adds $432
region_southeast	+$510	Low Impact	Southeast region slightly higher
region_northwest	-$267	Low Impact	Northwest region slightly lower
sex_male	-$496	Negligible	Minimal gender impact

7.2 Formula for Prediction

Predicted Charges = Intercept + (256 x age) + (332 x bmi) + (432 x children) + (23,612 x smoker) + (340 x age_smoker) + (380 x bmi_smoker) + (510 x region_southeast) - (267 x region_northwest) - (496 x sex_male)

Where:

- smoker = 1 if smoker, 0 if non-smoker
- age_smoker = age x smoker

- bmi_smoker = bmi x smoker

8. BUSINESS RECOMMENDATIONS

Recommendation 1: Implement Smoking-Based Pricing Structure

- Set baseline premium for non-smokers
- Apply $23,600+ surcharge for smokers
- Offer smoking cessation programs with premium reductions
- Expected ROI: High - smoking is the dominant factor

Recommendation 2: Develop Age-Based Sliding Scale

Age Group	Non-Smoker Premium	Smoker Premium
18-30	Baseline	Baseline + $23,600
31-40	Baseline + $2,560	Baseline + $23,600 + $3,400
41-50	Baseline + $5,120	Baseline + $23,600 + $6,800
51-64	Baseline + $8,960	Baseline + $23,600 + $11,900

Note: Age effect compounds with smoking status

Recommendation 3: Incentivize Healthy BMI

BMI Category	Recommended Action
Healthy (less than 25)	Offer premium discounts
Overweight (25-30)	No adjustment
Obese (30+)	Apply surcharge with wellness program option

Savings Potential: Each BMI point reduction saves approximately $330 annually

Recommendation 4: Simplify Regional Pricing

- Maintain uniform pricing
- Regional differences less than $500
- Avoid unnecessary complexity in pricing structure

- Focus resources on smoking cessation and wellness programs instead

9. LIMITATIONS AND FUTURE WORK

9.1 Current Limitations

Limitation	Impact
US-based dataset	May not generalize to South African market
Linear assumptions	Assumes linear relationships (validated through EDA)
Limited features	No medical history, chronic conditions, occupation
No time dimension	Cannot account for premium changes over time

9.2 Suggested Future Improvements

Improvement	Priority	Description
South African data collection	High	Collect local customer data for calibration
Add medical history features	Medium	Chronic conditions, family history, prior claims
Explore non-linear models	Medium	Random Forest, XGBoost for comparison
Cross-validation	Low	Implement k-fold for more robust evaluation
Feature engineering	Low	Create additional interaction terms

10. CONCLUSION

A linear regression model with interaction terms was successfully developed to predict medical insurance charges with 85% accuracy (R^2). The model identified smoking status as the dominant predictor, followed by age and BMI.

Key Takeaways for the Medical Aid Scheme:

- Smoking is the most significant factor - Smokers pay 300% more than non-smokers

- Age and BMI effects are amplified for smokers - Interaction terms improve model accuracy by 9%
- Regional differences are minimal - Uniform pricing is acceptable
- Gender has no meaningful impact - Pricing should not differentiate by gender

Model Readiness:

Aspect	Status
Technical Performance	Ready (85% R^2, $5,180 RMSE)
Business Interpretability	Ready (clear coefficients)
South African Context	Needs calibration
Deployment	Ready for pilot testing

The model provides a robust foundation for implementing a fair, data-driven pricing structure that appropriately accounts for customer lifestyles and demographics.

11. REFERENCES

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APPENDIX A: MODEL OUTPUT SUMMARY

Initial Model Metrics

Metric	Training	Testing
RMSE	$5,800	$6,062
MAE	$4,100	$4,183
R^2	0.79	0.78
MAPE	33.5%	35.2%

Improved Model Metrics

Metric	Training	Testing
RMSE	$5,100	$5,180
MAE	$3,500	$3,610
R^2	0.86	0.85
MAPE	26.8%	28.1%

END OF REPORT!!!!!